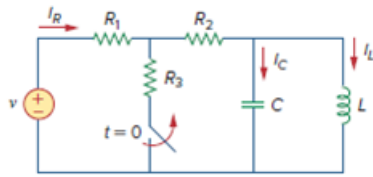


Problem

Using Fig. 8.63, design a problem to help other students better understand finding initial and final values.

Figure 8.63:

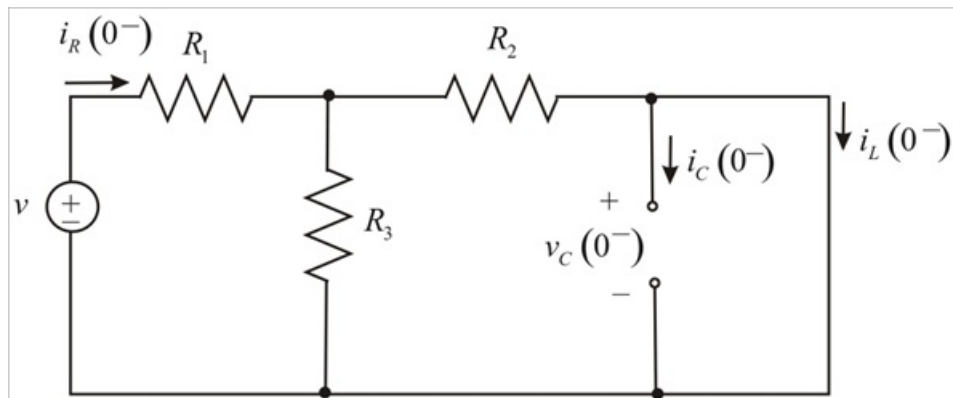


Step-by-step solution

Step 1 of 9

Refer to the circuit diagram in Figure 8.63 in the textbook.

For $t < 0$, the switch is closed for a long time, therefore the circuit has reached steady state and the capacitor can be replaced by an open circuit and the inductor by a short circuit,



Step 2 of 9

The equivalent resistance is,

$$R_{eq} = R_1 + (R_2 \parallel R_3)$$

$$R_{eq} = R_1 + \left(\frac{R_2 \times R_3}{R_2 + R_3} \right)$$

$$R_{eq} = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2 + R_3}$$

Step 3 of 9

The current $i_R(0^-)$ is,

$$i_R(0^-) = \frac{v}{R_{eq}}$$

$$i_R(0^-) = \frac{R_2 + R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3} v$$

Due to the open circuit, current through capacitor is zero.

$$i_C(0^-) = 0 \text{ A}$$

Step 4 of 9

By current division rule,

$$i_L(0^-) = i_R(0^-) \left[\frac{R_3}{R_3 + R_2} \right]$$

$$i_L(0^-) = \frac{v(R_2 + R_3)}{R_1 R_2 + R_2 R_3 + R_1 R_3} \left[\frac{R_3}{R_2 + R_3} \right]$$

$$i_L(0^-) = \frac{R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3} v$$

Step 5 of 9

Since the capacitor is in parallel with the inductor,

$$v_C(0^-) = 0 \text{ V}$$

Since the capacitor voltage and inductor current cannot change abruptly,

$$i_L(0^+) = i_L(0^-)$$

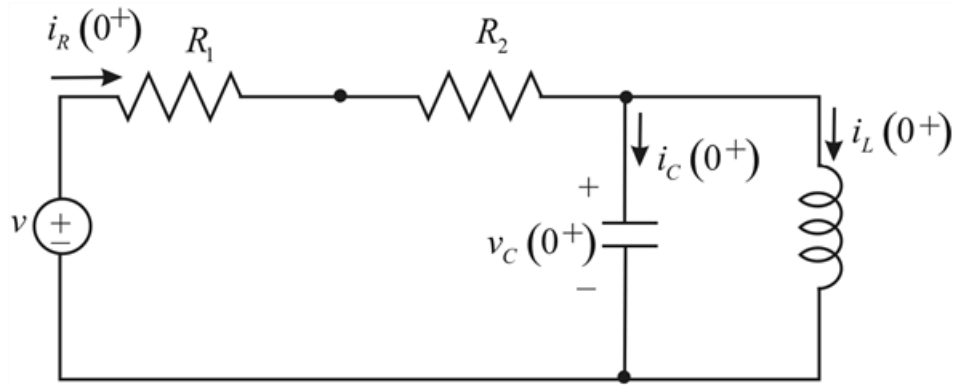
$$v_C(0^+) = v_C(0^-)$$

$$i_L(0^+) = \frac{R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3} v$$

$$v_C(0^+) = 0 \text{ V}$$

Step 6 of 9

For $t=0^+$, the switch is opened; therefore the equivalent circuit is,



Since $v_C(0^+)=0V$ (i.e. short circuit)

$$i_R(0^+) = \frac{v - v_C(0^+)}{R_1 + R_2}$$

$$= \frac{v}{R_1 + R_2}$$

Step 7 of 9

Write Kirchhoff's current law at the node.

$$-i_R(0^+) + i_C(0^+) + i_L(0^+) = 0$$

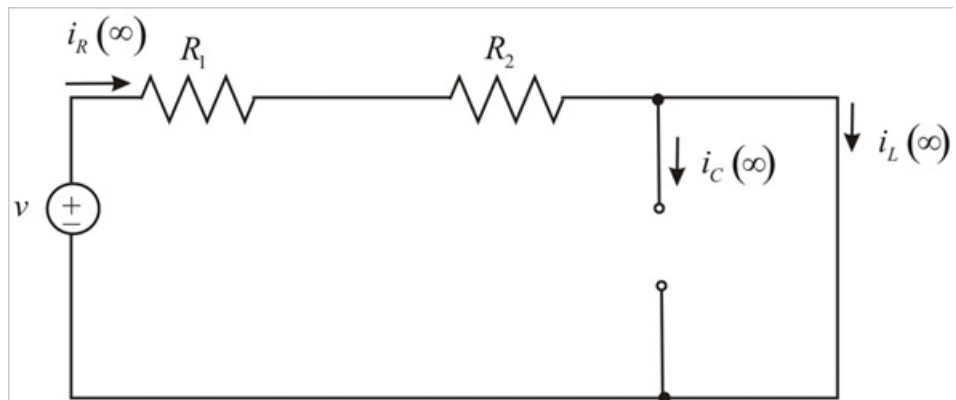
$$i_C(0^+) = i_R(0^+) - i_L(0^+)$$

$$i_C(0^+) = \frac{v}{R_1 + R_2} - \frac{R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3} v$$

$$i_C(0^+) = v \left[\frac{1}{R_1 + R_2} - \frac{R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3} \right]$$

Step 8 of 9

As $t \rightarrow \infty$, the circuit reaches steady state and the capacitor is replaced by an open circuit and the inductor is replaced by a short circuit,



Step 9 of 9

Due to open circuit,

$$i_C(\infty) = 0 \text{ A}$$

Since it is a series circuit, $i_R(\infty) = i_L(\infty)$

$$i_R(\infty) = i_L(\infty) = \frac{v}{R_1 + R_2}$$

The capacitor voltage $v_C(\infty)$ is 0 V .